

Introduction to Dynamic RoutingTheory

Task 1 :- List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

ANS :-

Feature	Static Routing	Dynamic Routing
How path is set	Fixed & Manual. An engineer types the route by hand.  <i>Analogy:</i> Using a printed paper map.	Automatic. Routers calculate the best route on their own.  <i>Analogy:</i> Google Maps auto-selecting the fastest route.
Handling roadblocks	Stuck. If a path breaks, data stops until fixed manually.  <i>Analogy:</i> A driver getting stuck at a closed road because they only have a paper map	Instant Reroute. Automatically finds an open detour.  <i>Analogy:</i> Swiggy rerouting a rider around bad traffic in real time.
Network Size	Best for tiny setups. Painful to update in big networks.  <i>Analogy:</i> Writing down directions by hand for 2 delivery stops vs. 1,000 stops.	Best for large setups. Scales easily to huge networks.  <i>Analogy:</i> Google Maps guiding millions of drivers worldwide simultaneously.

Task 2 :- Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

ANS :-

Open the File: Double-click your saved Packet Tracer network file.

Click Router 1: Go to the CLI tab.

Type these commands:

- show ip route (*Look for lines starting with R—these are RIP routes*).
- show ip protocols (*Check RIP timer details*).

Repeat: Do the exact same thing on Router 2 and Router 3.

Write down:

- The destination networks marked with R.
- The metric/hop count (the number inside the brackets, e.g., [120/1] → 1 hop).

Task 3 :- Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.

ANS :-

What to Do

1. Pick a router in your network and shut down one of its connected interfaces (link).
2. Start a timer immediately.

What to Observe & Record

- Holddown Timer (180 sec): The neighboring router won't delete the route immediately. It marks it as "possibly down" and waits to see if it comes back.
- Flush Timer (240 sec total): After 4 minutes (240 seconds) total, the unreachable route is completely deleted from the routing table.

Why it takes so long

RIP is slow to react to network changes because it relies on standard periodic timers instead of instant updates.

Task 4 :- Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.

ANS :-

Both Uber or Zomato and RIP routing constantly check for path changes and update automatically without human intervention. When a road is blocked or a network link fails, both systems automatically discover a new alternative path and update their directions. However, just like RIP sends periodic updates every 30 seconds to refresh its table, delivery apps continually refresh map to show the newest live route